

Technical Operations Manual

1. System Overview

The application consists of a FastAPI service, a PostgreSQL database, background workers, and a reverse proxy. Production services are deployed using containers.

2. Deployment

Deployments should be performed through the approved CI/CD pipeline. Changes must pass automated tests before production deployment.

3. Health Checks

The application exposes a health endpoint that verifies service availability. Database connectivity should also be checked during operational diagnostics.

4. Logging

Application logs should include timestamps, log levels, request identifiers, and useful error context. Secrets and credentials must never be written to logs.

5. Incident Response

When an incident occurs, first determine the affected component and user impact. Review application logs, recent deployments, dependency health, and database connectivity before making corrective changes.

6. Database Backup

Production database backups should run according to the organization's backup schedule. Restore procedures should be tested periodically rather than assuming that backups are usable.

7. Security

API credentials must be stored using environment variables or an approved secret-management system. Access should follow the principle of least privilege.

8. Rollback

If a deployment causes a significant production issue, use the approved rollback procedure and record the incident, root cause, and corrective action.